Composition Order and Inverse Domains
Answer Key

Algebra 2 — Error Analysis

Quick Answers

1. 15	3. 5	5. x	7. 3, -3
2. 2	4. -5	6. 7	8. —

Curriculum

Level: Algebra 2 — Error Analysis
HSF-BF.A.1c, HSF-BF.B.4 – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–5 of 5 across 8 tagged checks.

Common wrong answers

4. *If they got 7: composed in the reverse order, computing g(f(4)) instead of f(g(4))*
6. *If they got -1: took the minus branch of the square root, which lands outside the domain*

1. $f(x) = 2x - 5$, $g(x) = x^2 + 1$; find $(f \circ g)(3)$.
Step 1. $(f \circ g)(3)$ means $f\big(g(3)\big)$ — the function written on the *right* acts first.
Step 2. $g(3) = 3^2 + 1 = 10$.
Step 3. $f(10) = 2(10) - 5 = 15$. $\boxed{15}$

2. Same f and g; find $(g \circ f)(3)$.
Step 1. Now f acts first: $(g \circ f)(3) = g\big(f(3)\big)$.
Step 2. $f(3) = 2(3) - 5 = 1$.
Step 3. $g(1) = 1^2 + 1 = 2$.
Comparison to state: $15 \neq 2$, so composition is *not* commutative — swapping the order changes the answer, and one counterexample settles it. $\boxed{2}$

3. $f(x) = 3x + 7$; find $f^{-1}(22)$.
Step 1. $f^{-1}(22)$ is the input whose output is 22, so solve $3x+7 = 22$ rather than substituting 22 into f.
Step 2. $3x = 15$, so $x = 5$.
Step 3. Check forward: $f(5) = 15 + 7 = 22$ ✓ $\boxed{x = 5}$

4. Find and fix. $f(x) = 3x + 1$, $g(x) = x - 6$; $(f \circ g)(4)$.
Step 1. Nia's first move was $f(4) = 13$, then she subtracted 6. That is $g\big(f(4)\big) = (g \circ f)(4) = 7$ — the reverse composition.
Step 2. The correct order feeds 4 to g first: $g(4) = 4 - 6 = -2$.
Step 3. Then $f(-2) = 3(-2) + 1 = -5$.
One-sentence diagnosis to accept: "She applied f first; in $f \circ g$ the inner function g is applied first." $\boxed{-5}$

5. $f(x) = 3x + 7$, $f^{-1}(x) = \dfrac{x-7}{3}$; show $f\big(f^{-1}(x)\big) = x$.
Step 1. Substitute the whole inverse expression into f: $f\big(f^{-1}(x)\big) = 3 \cdot \dfrac{x-7}{3} + 7$.
Step 2. The 3 and the denominator 3 cancel: $(x - 7) + 7$.
Step 3. $x - 7 + 7 = x$ for every real x, which is exactly the identity that certifies an inverse. $\boxed{f(f^{-1}(x)) = x}$

6. Find and fix. $h(x) = (x - 3)^2$ on $x \geq 3$; Sam's $h^{-1}(x) = 3 - \sqrt{x}$ gives $h^{-1}(16) = -1$.
Step 1. An inverse must return values *inside the domain of* h, and $-1 \geq 3$ is false, so Sam's output cannot be an input of h.
Step 2. Build the rule honestly: from $y = (x - 3)^2$, $\sqrt{y} = |x - 3| = x - 3$ because $x \geq 3$ makes $x - 3$ non-negative. So $h^{-1}(x) = 3 + \sqrt{x}$ — the plus branch is forced by the restriction.
Step 3. $h^{-1}(16) = 3 + 4 = 7$, and checking forward, $h(7) = (7 - 3)^2 = 16$ ✓
Diagnosis to accept: "He kept the minus branch of the square root, which leaves the restricted domain $x \geq 3$." $\boxed{7}$

7. $g(x) = x^2$ on all of $\mathbb{R}$; solve $x^2 = 9$.
Step 1. $x^2 - 9 = 0$ factors as $(x - 3)(x + 3) = 0$, so the equation has two real solutions, not one.
Step 2. $x = 3$ and $x = -3$ both square to 9.
What it proves: two different inputs share the output 9, so g is not one-to-one on $\mathbb{R}$ and has no inverse there. Writing $g^{-1}(x) = \sqrt{x}$ silently keeps only the $x \geq 0$ half of the domain; the claim is legitimate only after g is restricted to $x \geq 0$. $\boxed{x = 3 \text{ or } x = -3}$

8. $f(x) = \sqrt{x - 1}$, $g(x) = x^2 + 1$; Marcus's claim.
Step 1. The step $\sqrt{x^2} = x$ is the flaw: $\sqrt{x^2} = |x|$, which equals x only when $x \geq 0$. For $x = -4$, $f(g(-4)) = \sqrt{17 - 1} = 4$, not -4.
Step 2. He checked only one direction. The second check, $g\big(f(x)\big) = (\sqrt{x - 1})^2 + 1 = x$, needs $x - 1 \geq 0$, i.e. $x \geq 1$ — and that is the restriction his one-sided check hid.
Step 3. Domains that make the pair genuine inverses: f has domain $x \geq 1$ with range $y \geq 0$; g must be restricted to $x \geq 0$, where its range is $y \geq 1$. Each function's domain is the other's range, which is the structural signature of an inverse pair.
Model answer (accept any equivalent wording): "$\sqrt{x^2} = |x|$, so $f(g(x)) = x$ only for $x \geq 0$. He never checked $g(f(x))$, which forces $x \geq 1$. With $f : x \geq 1 \to y \geq 0$ and $g : x \geq 0 \to y \geq 1$ the two are genuine inverses."